

Saif E. Nouma, PhD

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RESEARCH INTERESTS

My research interests include **applied cryptography, network security, IoT security, and machine learning**, with a focus on advancing the security and efficiency of emerging computing platforms. I am also interested in the security and efficiency of digital twins using post-quantum cryptography, edge computing, and distributed designs.

EDUCATION

Doctor of Philosophy in Computer Science University of South Florida ◦ GPA: 3.9/4.0 ◦ Advisor: Dr. Attila Altay Yavuz ◦ Thesis: Lightweight and Resilient Cryptographic Protocols for Internet of Things	Aug 2021-May 2026 Tampa, FL, USA
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Bachelor of Science in Computer Science University of Carthage ◦ Advisor: Dr. Khalil Drira ◦ Thesis: Applications of Machine Learning in Networking and IoT	Aug 2017-Jan 2020 Tunis, Tunisia
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Associate of Science in Mathematics University of Monastir	Aug 2015-Jun 2017 Monastir, Tunisia
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EMPLOYMENT HISTORY

Graduate Research Assistant University of South Florida, Tampa, FL, USA	Aug 2021-May 2026
Graduate Teaching Assistant University of South Florida, Tampa, FL, USA	Aug 2021-May 2026
System Administrator University of South Florida, Tampa, FL, USA	Aug 2023-Jun 2024
Software Engineer Kopileft Services Inc., Tunis, Tunisia	Jul 2020-Jul 2021
ML Engineer Intern LAAS-CNRS, France, Toulouse	Jan 2020-Jul 2020
ML Engineer Intern Wevioo, Tunis, Tunisia	Jun 2019-Aug 2019

PATENTS & PUBLICATIONS

PATENT

- [P2] Attila A. Yavuz, **Saif E. Nouma**. System and Method for Cryptographic Forensic Audits on Lightweight IoT and Digital Archives. **US Patent** US20240007300A1, 2024.
- [P1] Attila A. Yavuz, **Saif E. Nouma**. Hardware Supported Authentication and Signatures for Wireless, Distributed and Blockchain Systems. **US Patent** US20230308289A1, 2023.

JOURNALS

- [J6] **Saif E. Nouma**, Attila A. Yavuz. Lightweight and High-Throughput Secure Logging for Internet of Things and Cold Cloud Continuum. *ACM Transactions on Internet of Things (TIIoT)*, 2026. (*Impact factor: 3.5*)
- [J5] **Saif E. Nouma**, Attila A. Yavuz. Lightweight and Resilient Signatures for Cloud-Assisted Embedded IoT Systems. *Wiley Security and Privacy*, Vol. 9, 2026. (*Impact factor: 2.1*)
- [J4] Attila A. Yavuz, Saleh Darzi, **Saif E. Nouma**. LiteQSign: Lightweight and Quantum-Safe Signatures for Heterogeneous IoT Applications. *IEEE Access*, Vol. 13, pp. 171442-171456 2025. (*Impact factor: 3.6*)

- [J3] Aaron Pendino, Nghia Nguyen, **Saif E. Nouma**, Jing Wang, Attila A. Yavuz, Yasin Yilmaz, Gokhan Mumcu. (2025). Additively Manufactured RF Electronics with Structurally Integrated Physically Unclonable Functions for Wireless System Security. *IEEE Access*, 2025. (Impact factor: 3.6)
- [J2] Kiarash Sedghighadikolaei, Attila A. Yavuz, **Saif E. Nouma**. Signer-Optimal Multiple-Time Post-Quantum Hash-Based Signature for Heterogeneous IoT Systems. *Internet of Things*, 2025. (Impact factor: 7.6)
- [J1] **Saif E. Nouma**, Attila A. Yavuz. Post-Quantum Hybrid Digital Signatures with Hardware-Support for Digital Twins. *ACM Transactions on Multimedia Computing, Communications, and Applications (ACM TOMM)*, 2024. (Impact factor: 6.0)

CONFERENCES

- [C7] **Saif E. Nouma**, Attila A. Yavuz. Lightweight and Breach-Resilient Authenticated Encryption Framework for Internet of Things. In *43rd IEEE Military Communications Conference (IEEE MILCOM)*, 2025.
- [C6] **Saif E. Nouma**, Attila A. Yavuz. Practical Cryptographic Forensic Tools for Lightweight Internet of Things and Cold Storage Systems. In *Proceedings of the 8th ACM/IEEE Conference on Internet of Things Design and Implementation (ACM/IEEE IoTDI)*, 2023.
- [C5] **Saif E. Nouma**, Attila A. Yavuz. Post-Quantum Forward-Secure Signatures with Hardware-Support for Internet of Things. In *58th IEEE International Conference on Communications (IEEE ICC)*, 2023.
- [C4] **Saif E. Nouma**, Attila A. Yavuz. Lightweight Digital Signatures for Internet of Things: Current and Post-Quantum Trends and Visions. In *6th IEEE Conference on Dependable and Secure Computing (DSC)*, 2023.
- [C3] Attila A. Yavuz, Kiarash Sedghighadikolaei, Saleh Darzi, **Saif E. Nouma**. Beyond Basic Trust: Envisioning the Future of NextGen Networked Systems and Digital Signatures. In *5th IEEE Conference on Trust, Privacy and Security in Intelligent Systems and Applications (IEEE TPS-ISA)*, 2023.
- [C2] Attila A. Yavuz, **Saif E. Nouma**, Thang Hoang, Duncan Earl, Scott Packard. Distributed Cyber infrastructures and Artificial Intelligence in Hybrid Post-Quantum Era. In *4th IEEE Conference on Trust, Privacy and Security in Intelligent Systems and Applications (IEEE TPS-ISA)*, 2022.
- [C1] Attila A. Yavuz, Duncan Earl, Scott Packard, **Saif E. Nouma**. Hybrid Low-Cost Quantum-Safe Key Distribution. In *Quantum 2.0 – Optica*, May 2022, MA, USA.

E-PRINTS

- [E3] **Saif E. Nouma**, Gokhan Mumcu, Attila A. Yavuz. Diamond: Design and Implementation of Breach-Resilient Authenticated Encryption Framework For Internet of Things, 2026. *Under review at ACM Transactions on Embedded Computing Systems (ACM TECS)*. (Impact factor: 2.8)
- [E2] Saleh Darzi, **Saif E. Nouma**, Kiarash Sedghi, Attila A. Yavuz. QPADL: Post-Quantum Private Spectrum Access with Verified Location and DoS Resilience. *arXiv preprint arXiv:2510.03631*, 2025. *Under review at IEEE Transactions on Information Forensics and Security (IEEE TIFS)*. (Impact factor: 8.0).
- [E1] **Saif E. Nouma**. (2020). Applications of Machine Learning (ML) in Networking and IoT. hal-02932494.

TEACHING EXPERIENCE

Graduate Teaching Assistant

- CIS 4212/6214: Privacy-Preserving and Trustworthy Cyber-Infrastructures Spring 2026
- COP 4538 IT Data Structures Fall 2021

Guest Lecturer

- CIS 4930/6930: Cryptography: Theory and Practice Fall 2025
- CIS 4212/6214: Privacy-Preserving and Trustworthy Cyber-Infrastructures Spring 2024

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2021 - May 2026

University of South Florida, Tampa, FL

I designed and implemented efficient authenticated encryption primitives, including:

- Graphene: Designed efficient and compromise-resilient authenticated encryption constructions based on AES-GCM and Chacha20-Poly1305 standards, with full-fledged implementation on 32-bit ARM Cortex-M4, achieving orders-of-

magnitude performance improvements and low-latency compared to the state-of-the-art counterparts, presented at IEEE MILCOM 25. My implementation is open-source at <https://github.com/saifnouma/graphene>

- Diamond: Improved the Graphene scheme by incorporating an efficient key update and more algorithmic formality with provable security. Extensive performance evaluation on embedded devices, 64-bit ARM Cortex-A72 and 8-bit AVR ATmega2560 confirms the efficiency of Diamond over Graphene and NIST lightweight cryptographic standard Ascon. My implementation is open-source at <https://github.com/saifnouma/diamond>

I designed and implemented efficient digital signature primitives for constrained devices, including:

- LRSHA: Designed and implemented an efficient and resilient digital signatures by leveraging distributed cloud assistance and secure hardware Intel SGX. Extensive performance evaluation on 8-bit AVR ATmega2560 and commodity hardware confirms LRSHA's efficiency. My implementation is open-source at <https://github.com/saifnouma/lrsha>
- POSLO: Designed efficient secure logging protocol, achieving extended battery longevity on constrained devices. Developed a novel GPU-accelerated algorithm for batch verification of large logs on the cloud, resulting in significant speedup compared to baseline CPU. Our work resulted in filing one US patent and publications at ACM/IEEE IoTDI 23 and ACM TIoT 25. My implementation is open-source at <https://github.com/saifnouma/poslo>
- HYHASES: Designed efficient, forward-secure, and post-quantum digital signatures with TEE hardware support, implemented on 8-bit AVR ATmega2560 and Intel SGX, achieving 34× better efficiency than NIST ML-DSA standard, presented at IEEE ICC 23. I later extended this work to a hybrid signature, leveraging a conventional aggregate signature to significantly reduce bandwidth overhead and energy usage on low-powered devices. The extended work is published in ACM TOMM 2024. My implementation is open-source at <https://github.com/saifnouma/hyhases>

PROFESSIONAL EXPERIENCE

System Administrator

Aug 2023-Jun 2024

University of South Florida, Tampa, FL

- Developed and configured real-time dashboards and alert rules using Prometheus and Grafana stacks, tracking I/O network throughput and node/container health across an HPC cluster and the machines of the college infrastructure
- Developed Ansible roles and playbooks to support continuous integration and task automation

Software Engineer

Jul 2020-Jul 2021

Kopileft Services Inc., Tunis, Tunisia

- Improved e-commerce backends by migrating Java-based SOAP web services to Kotlin, developed services on Apache Tomcat, and implemented Exposed-based data-access layers with PostgreSQL
- Implemented the BI reporting and dashboard infrastructure by optimizing $\approx 50\%$ of SQL queries and stored procedures written in PostgreSQL, improving data freshness and reliability

ML Engineer Intern

Jan 2020-Jul 2020

LAAS-CNRS, Toulouse, France

- Implemented and benchmarked LSTM/GRU models using TensorFlow and scikit-learn on a real-world network traffic (1.6 TB capture), achieving 3% MAPE and halving error compared to SVR. My implementation is open-source at <https://github.com/SaifNOUMA/Network-Traffic-Prediction>
- Implemented a distributed early-exit CNN model for edge IoT networks using TensorFlow, containerized with Docker, and performed communication using ZeroMQ, enabling faster inference. My implementation is open-source at https://github.com/SaifNOUMA/Edge-Computing/tree/master/dnn_partitioning
- Developed and implemented a DQN algorithm for dynamic task offloading and resource allocation in multi-user edge computing, using ns3-gym, achieving improved latency-energy tradeoffs. My implementation is open-source at https://github.com/SaifNOUMA/Edge-Computing/tree/master/task_offloading_optimization

ML Engineer Intern

Jun 2019-Aug 2019

Wevioo, Tunis, Tunisia

- Developed, trained, and fine-tuned a Siamese CNN (S-CNN) with contrastive loss on a collected real-world dataset of genuine/forged handwritten signature pairs for fake signature detection
- Collaborated with computer-vision team on signature extraction from handwritten bank checks and deployed the prototype to local banks, which served as a major step towards mitigating fraud

SKILLS	
Programming Languages: C/C++, Assembly, Python, Java, Kotlin, MATLAB, CUDA	
Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib	
Security & Cryotography: OpenSSL, WolfSSL, Mbed-TLS, MITRE ATT&CK, SGX	
DevOps: Git, SVN, Docker, Ansible, Prometheus, Grafana, Nagios, Slurm	
Database: PostgreSQL, MySQL	
Embedded Hardware: ARM Cortex-M4, ARM Cortex-A72, 8-bit AVR ATmega series	
HONORS AND AWARDS	
NSF/USF International Travel Grant for IEEE ICC 2023 (\$3,000)	2023
Top 0.5% (Rank 4/800) at Tunisian National Engineering Entrance Exam	2017
PUBLIC TALKS	
Research paper presentation at IEEE MILCOM, Los Angeles, CA, USA	2025
PhD major area Presentation at Tampa, FL, USA	2024
Research paper presentation at IEEE DSC, Tampa, FL, USA	2023
Research paper presentation at ACM/IEEE IoTDI, San Antonio, TX, USA	2023
SERVICES	
Reviewer	
• IEEE Transactions on Dependable and Secure Computing (IEEE TDSC), IEEE Transactions on Mobile Computing (IEEE TMC), Scientific Reports, Cluster Computing	2026
• Blockchain: Research and Applications (Elsevier), Computer Networks (Elsevier), Reliability Engineering and System Safety (Elsevier), Security and Privacy (Wiley)	2025
• IEEE Transactions on Information Forensics and Security (IEEE TIFS)	2024
REFERENCES	
References are available upon request	